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Department of Computer Engineering and Applications
Institute of Engineering & Technology

BCSE-1501
Network Security

Unit-2 Network Security And Protocols

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CIA TRAID

Confidentiality

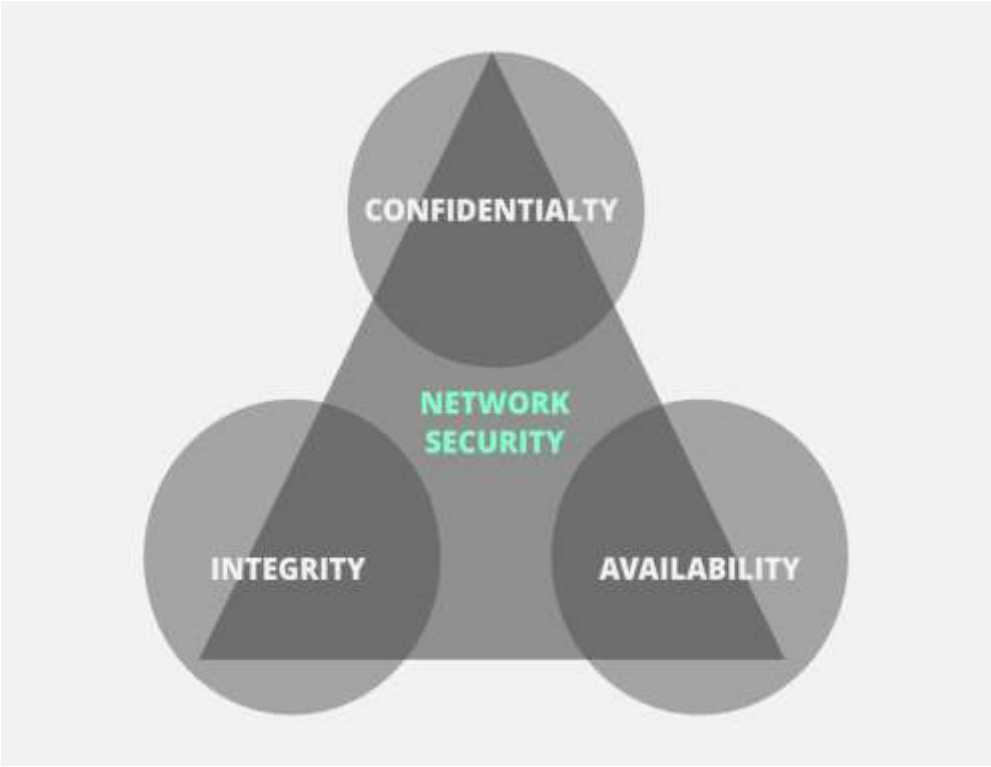
•Ensures data is accessed only by authorized users
•Prevents data leakage and eavesdropping
•Achieved using:
- Encryption (AES, RSA)
- Authentication mechanisms
- Access control lists (ACLs)

Integrity

•Ensures data is accurate and not altered in transit
•Prevents unauthorized modification
•Achieved using:
- Hash functions (SHA)
- Digital signatures
- Message authentication codes (MAC)

Availability

•Ensures network services are accessible when needed
•Protects against service disruption
•Achieved using:
- Redundancy and load balancing
- Backups
- DoS/DDoS protection



Network Services & Common Ports

What Are Network Ports?

- A **port** is a logical communication endpoint used by transport layer protocols (**TCP/UDP**)
- Port numbers range from **0–65535**
- Used to identify **specific services/applications** on a host

Port Categories:

Category	Range	Description
Well-known ports	0–1023	Standard services (IANA assigned)
Registered ports	1024–49151	Vendor/user applications
Dynamic/Private ports	49152–65535	Temporary client-side ports

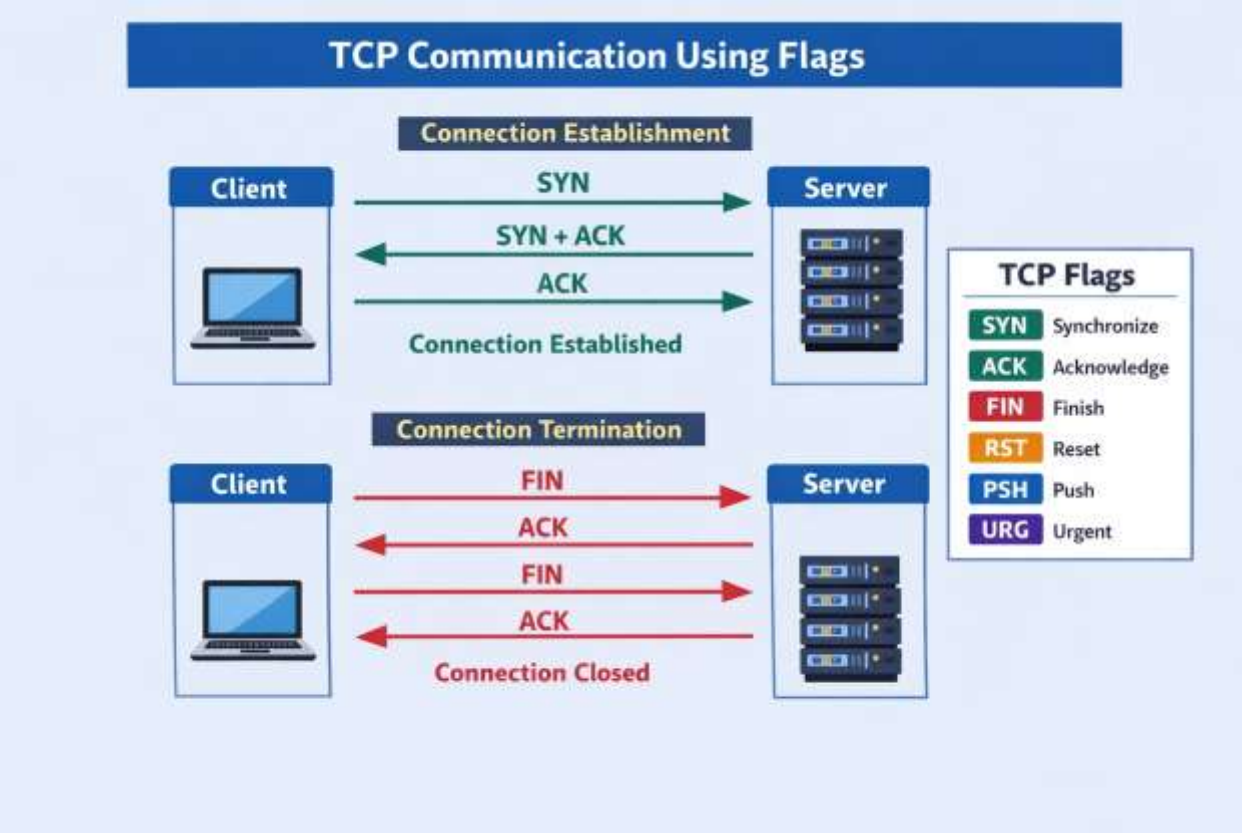
Port	Service	Protocol	Purpose
21	FTP	TCP	File transfer
22	SSH / SFTP	TCP	Secure remote login
23	Telnet	TCP	Remote login
25	SMTP	TCP	Send emails
53	DNS	TCP/UDP	Name resolution
69	TFTP	UDP	Simple file transfer
80	HTTP	TCP	Web communication
110	POP3	TCP	Receive emails
143	IMAP	TCP	Email access
443	HTTPS	TCP	Secure web traffic
587	SMTP Submission	TCP	Secure email sending
8080	HTTP (Alternate)	TCP	Proxy / testing

Port	Service	Protocol	Security Relevance
123	NTP	UDP	Amplification attacks
139	NetBIOS	TCP	Legacy Windows sharing
161	SNMP	UDP	Weak default configs
445	SMB	TCP	Ransomware target
1433	MS SQL Server	TCP	Brute-force attacks
3306	MySQL	TCP	Exposed databases
3389	RDP	TCP/UDP	Brute-force
5900	VNC	TCP	Unauthorized access

What are TCP Flags?

TCP flags are **control bits** in the TCP header that manage **connection establishment, data transfer, and termination** between communicating hosts.

- Each flag is **1 bit**
- Used to control **state and behavior** of TCP communication
- Help detect **errors, abnormal behavior, and attacks**



Common TCP Flags and Their Functions

Flag	Full Form	Purpose / Function	Security Relevance
SYN	Synchronize	Initiates TCP connection	Used in SYN flood
ACK	Acknowledgment	Confirms receipt of data	Abnormal ACKs indicate attacks
FIN	Finish	Graceful connection termination	Normal session end
RST	Reset	Abruptly terminates connection	Session hijacking
PSH	Push	Sends data immediately	Data exfiltration
URG	Urgent	Marks urgent data	Rare, misuse possible

Port Scanning with Nmap

❖**What is Port Scanning?**

- Port scanning is the process of **identifying open, closed, and filtered ports** on a target system to determine:
- Running services
- Service versions
- Possible vulnerabilities
- It is the **first step of reconnaissance** in penetration testing and cyber attacks.

❖**Why Port Scanning is Important**

- Identifies **open ports** (attack surface)
- Detects **unauthorized services**
- Helps in **vulnerability assessment**
- Supports **incident response & forensics**

Common Nmap Scan Types

Scan Type	Nmap Option	Purpose / Description
TCP Connect Scan	`-sT`	Completes full TCP handshake; easily detected
SYN (Stealth) Scan	`-sS`	Half-open scan; fast and stealthy
UDP Scan	`-sU`	Scans UDP services (DNS, SNMP)
FIN Scan	`-sF`	Firewall evasion using FIN flag
NULL Scan	`-sN`	Sends packets with no flags
XMAS Scan	`-sX`	FIN+PSH+URG flags set
ACK Scan	`-sA`	Detects firewall rules
Window Scan	`-sW`	Determines firewall behavior
Maimon Scan	`-sM`	Advanced FIN-based scan
Idle (Zombie) Scan	`-sI`	Highly stealth scan using zombie host
IP Protocol Scan	`-sO`	Detects supported IP protocols
Ping Scan	`-sn`	Discovers live hosts only
Service Version Scan	`-sV`	Detects service versions
OS Detection	`-O`	Identifies target operating system

Attacks Across Network Layers – DATA Link Layer

Link Layer (Data Link Layer)
Responsible for **MAC addressing** and **local network communication** within a LAN.

Common Attacks

- **ARP Spoofing / ARP Poisoning:**
  Attacker sends fake ARP messages to associate their MAC with a victim's IP → enables **Man-in-the-Middle attacks**.
- **MAC Flooding:**
  Overloads switch CAM tables with fake MAC addresses → causes switch to behave like a hub → allows **traffic sniffing**.
- **Packet Sniffing:**
  Captures unencrypted LAN traffic → exposes sensitive **credentials and data**.
- **VLAN Hopping:**
  Exploits switch misconfigurations to access **unauthorized VLANs** → bypasses network segmentation.

Impact

- Interception of network traffic
- Data and credential theft
- Network performance degradation
- Unauthorized access

Network layers attacks

The **Network Layer** handles IP addressing and routing. Attacks at this layer target IP packets and routing mechanisms.

1. IP Spoofing

The attacker forges the source IP address to impersonate a trusted system and bypass security controls.

2. Packet Sniffing

Unauthorized capturing of IP packets to obtain sensitive information transmitted over the network.

3. Routing Attacks

Manipulation of routing protocols (RIP, OSPF, BGP) to redirect, intercept, or drop network traffic.

4. ICMP Attacks

Misuse of ICMP messages such as ICMP flood and ping of death to disrupt network communication.

5. Man-in-the-Middle (MITM)

The attacker intercepts and alters IP packets between communicating hosts without their knowledge.

Transport Layer Attacks

- Provide end to end communication between hosts.
- Ensure Data Segmentation
- Manages **port numbers** for process-to-process delivery
- Provides **flow control and error control** (TCP)

Attacks on Transport layer:

- **TCP Reset Attack** – Terminates active TCP connections using fake RST packets
- **UDP Flood** – Overwhelms the target with excessive UDP traffic
- **Replay Attack** – Reuses captured packets to gain unauthorized access
- **TCP SYN Flood** – Exhausts server resources with half-open connections
- **Session Hijacking** – Takes control of an active TCP session
- **Port Scanning** – Identifies open ports and running services

Application Layer Attacks

- Enables **interaction between users and applications**
- Provides services like **HTTP, FTP, SMTP, DNS**
- Handles **request–response communication** between applications

Application Layer Attacks

- **SQL Injection** – Attacker inserts malicious SQL queries through input fields to access or modify database data.
- **Cross-Site Scripting (XSS)** – Malicious scripts are injected into web pages and executed in the user's browser.
- **Cross-Site Request Forgery (CSRF)** – Forces an authenticated user to perform unwanted actions without consent.
- **Malware / Ransomware** – Malicious software that damages systems or encrypts data for ransom.
- **Phishing** – Fake emails or websites trick users into revealing sensitive information.
- **Application-layer DoS** – Excessive application requests overload servers, making services unavailable

Security countermeasures

- **Security countermeasures** are techniques used to **prevent, detect, and respond** to security threats.
- **Key Security Countermeasures**
- **Firewalls** – Filter unauthorized network traffic
- **Encryption** – Protects data confidentiality
- **Authentication & Authorization** – Verifies users and controls access
- **IDS / IPS** – Detects and prevents intrusions
- **Access Control Policies** – Restrict resource usage
- **Regular Updates & Patching** – Fix known vulnerabilities

Network Access Control (NAC)

- Network Access Control (NAC) is a security mechanism that controls who can access a network and what level of access they are given. It ensures that only authorized users and secure devices are allowed to connect to the network.

How NAC Works

- When a user or device tries to connect to a network, NAC:
- Authenticates the user or device (checks identity)
- Authorizes access based on role or policy
- Checks device compliance (OS updates, antivirus, firewall)
- If the device is not secure, NAC can restrict or quarantine it until security requirements are met.

Key Components of NAC

- **Authentication** – Verifies identity (username, password, certificates)
- **Authorization** – Grants role-based access
- **Accounting** – Logs user and device activity

Benefits of NAC

- Prevents unauthorized network access
- Reduces internal threats
- Improves network visibility
- Supports Zero Trust security model